



UNIVERSITY OF
CHEMISTRY AND TECHNOLOGY
PRAGUE

Faculty of Chemical Engineering

Computational Strategies for Nonadiabatic Molecular Dynamics

DISSERTATION

AUTHOR

SUPERVISOR

PROGRAMME

YEAR

Ing. Tomáš Jíra

prof. RNDr. Bc. Petr Slavíček,

Ph.D.

Chemistry

2026



VYSOKÁ ŠKOLA
CHEMICKO-TECHNOLOGICKÁ
V PRAZE

Fakulta chemicko-inženýrská

Výpočetní strategie pro neadiabatickou molekulární dynamiku

DISERTAČNÍ PRÁCE

VYPRACOVAL

ŠKOLITEL

PROGRAM

ROK

Ing. Tomáš Jíra

prof. RNDr. Bc. Petr Slavíček,

Ph.D.

Chemie

2026

This thesis was written at Department of Physical Chemistry of the University of Chemistry and Technology, Prague in September 2022—May 2026. This thesis was written at Department of Physical Chemistry of the University of Chemistry and Technology, Prague in September 2022—May 2026.

I hereby declare that this thesis is my own work. Where other sources of information have been used, they have been acknowledged and referenced in the list of used literature and other sources. I further declare that, in the event that a generative artificial intelligence model was used, this tool was employed in accordance with academic ethics; I have reviewed all content and assume full responsibility for it.

I have been informed that the rights and obligations implied by Act No. 121/2000 Coll. on Copyright, Rights Related to Copyright and on the Amendment of Certain Laws (Copyright Act) apply to my work. In particular, I am aware that the University of Chemistry and Technology in Prague, or another educational institution where this thesis was produced, has the right to sign a license agreement for use of this work as school work under Section 60 paragraph 1 of the Copyright Act. Should I in the future grant a licence to a third party for use of this work, the University of Chemistry and Technology in Prague, or another educational institution where this thesis was completed, may claim a reasonable contribution to cover the costs incurred in the creation of the work, even up to the full amount of such costs.

I agree to the publication of my work in accordance with Act No. 111/1998 Coll. on Higher Education, as amended.

In Prague on

.....
Ing. Tomáš Jíra